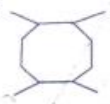
FEDERAL UNIVERSITY OF AGRICULTURE ABEOKUTA

CHM 102 PAST QUESTION

CHM 102

1. Blood sugar is another name for what? (a) Lactose (b) Fructose (c) glucose (d)
2. Carbohydrate have general formula of $(n(H_20)m$. which of the following compo its n = m? (a) Glycogen (b) Fructose (c) Glucose (d) Sucrose
3. Alkanes form a homologous series of _____ corresponding to general formular _____ (a) Saturated hydrocarbon / C_nH_{2n+2} (b) Saturated hydrocarbon / C_nH_{2n} (c) Unsaturated hydrocarbon / C_nH_{2n} (d) Unsaturated hydrocarbon / $C_n H_{2n}$
4. which of the following does not represent an alkane? (a) C_6H_4 (b) $C_{13}H_{28}$ (c) $C_{18}H_{36}$ (d) $C_{19}H_{40}$
5. how many straight chain alkanes correspond to the formular $C_4H_8C_{l2}$? (a) 3 (b) 6 (c) 4 (d) 8
6. In heterocyclic monosaccharides, a 5 ring compound is called (a) Pentanal (b) Pentanane (c) Furanose (d) Pyranose
7. Cl Br



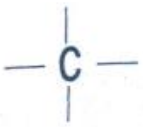
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F CH_3

(a) 1 bromo- 6chloro- fluro-3methylcyclobe peane (b) 4-bromo-6 chloro-1fluro-2methyl cycloheptane

(c)1-chloro-3bromo-5methyl-6flurocycloheptane (d) 3-bromo-1-chloro-6fluro-5methylcycloheptane

8. What is the approximate H-C-H in methane (a) 1090 (b) 1200 (c) 1800 (d) 1780
9. What is the IUPAC name for iso butane? (a) Isopropylmethane (b) 3-methylbutane (c) 2-methyl propane (d) 3-methylpentane.
10. A chiral carbon is a carbon having how many groups attached to it. (a)1 (b) 2 (c) 3 (d) 4



11. The process of producing ethane from sugar or starch is (a) Oxidation (b)Evaporation (c) Fermentation (d) Hydrogenation

12. Dehydration of alcohol at a temperature between <150°C yield (a) olefin (b) Alkanes (c) ether (d) Hydrogen

13. The stability of phenoxide ions is an indication of phenol being (a) Acidic (b) Basic (c) Monoprotic (d) Electrophlli

14. The structure RCHOH indicates _____ Alcohol (a) Primary (b) Secondary (c) Tertiary (d) Simple

15. The hybridized state of carbon in octene is ? (a) Sp (b) Sp^2 (c) Sp^3 (d) d^2Sp^2

16. Identify the functional groups in the compound below

CHCH- CHCOOH

NH_2 Ol

(a) Chloroamide and carboxylic acid groups (b) Chloroamine and carboxylic acid groups (c) Chloroamine, carbonylanohydroxyl groups (d) Chloroamide, carbonyl and hydroxyl groups

17. GABA is found in brain and is another name for what? (a) 4-Aminobutanoic acid (b) 2-Amino pentanoic acid (c)4-glucobutanoic acid (d) 2-Aminobutanoic acid.

18. Alcohol molecules are stabilized by _____ bond (a) Covalent (b)Ionic (c)Chemical (d) hydrogen

19. A dissacharride having D-Galacto Pyranose and D-Glycopyranose joined together by B-1, 4 glycosidic bind is called what? (a) Lactose (b) Maltose (c) Glucose (d) Sucrose

20. What is the name of the compound drawn below?



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(a)Pent-l-en-yne (b)Pent-4-en-1-yne (c) Pentene- I-yne (d) Penytne-1 ene

21. Example of an organic compound that have both covalent and ionic bond is _____ (a) Organometallic compounds (b) Amino acids (c) Carbohydrates (d) Fat and oil

22. Organic compounds generally lower m.pt and b.pt, due to (a) Weak hydrogen bonds (b) Presence of carbon atom (c) Weak fixes of all reaction (d) Solubility of this salts.

The following are components of natural gas except (a) Methane (b) Butane (c) Pro... (d) Propane

24. Polysaccharides are polymers of monosaccharide units obtained through polymerization (a) addition (b) Condensation (c) Step growth (d) Band c

25. An element with many unpaired electrons in ____ element (a) Paramagnetic (b) Diamagnetic (c) Ferromagnetic (d) Magnetic

26. When alkane contains three or more carbon atoms, substitution gives rise to (a) Polymerphism (b) Allotropy (c) Isomerism (d) Isotopy

27. The name given to cyclic structure of monosaccharide is _____ (a) Chirality (b) Fischer (c) Harwoth projection (d) Dextro & Larvorotatory

28. The carbon – carbon triple bonds consist of (a) 2r and 1TT (b) 1r and 2^ (c) 2r and 2^ (d) 1r and 1^

29. Sp3 hybridization can be found in which of the following organic compounds (a) Ethane (b) Ethene (c) Propyne (d) Allylic

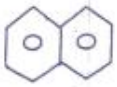
30. Thyroxine and triodothyroxine are the two hormones derive from _____ (a) Sugar (b) Lipid (c) Nucleic acid (d) Amino acid

31. Among the 20 common amino acids, how many have polar side chains (a) 9 (b) 5 (c) 4 (d) 3

32. _____ is a name given to short polymer of amino acids (a) Zwitterion (b) Primary amino acids (c) Peptide (d) Non - polar

33. A family of compound whose formular differ CH_2 is called _____ (a) Homologous series (b) Functional series (c) Halogen series (d) Aliphatic series

34. _____ is the most widely distributed plant skeletal polysaccharides. (a) Cellulose (b) Glycogen (c) Starch (d) Glucose

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The name of the compound is (a) Dibenzene (b) Dicyclohexane (c) Naphthalene (d) Phenylbenzene

36. Primary alcohol is completely oxidized to _____ (a) Alkanone (b) Carboxylic acid (c) Alkane (d) Alkene

(c) Pr

e original name of alkanes is paraffin meaning _____ (a) High affinity (b) Little affinity

(c) Electron affinity (d) Acidic affinity

_____ is a composed of continous branch chain of 3,500 d – glucose unit (a) Actin (b) Z amylase (c) Amylopectin (d) Myasin

39. Glycogen is found in which part of the body (a) Liver & Muscle (b) Liver & Urea (c) Kidney & Bile (d) Brain

40. The dehydrogenation of an alkane compound will lead to the formation of an organic compound _____ (a) Alcohol (b) Alkyne (c) Alkane (d) Alkene

41. What is the total number of sigma bonds found in the following compound? (a) CH_3-CH= C=CH- C=C-H

(a)8 (b) 10 (c) 11 (d) 15

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42. The simplest form of monosaccharides is known as _____ (a) Manose (b) Ethose (c) Tetrose (d) Triose

43. A carbon that bonded to 4 different groups is said to have (a)Larvo (b) Isomers (b) Chirality

(d) Dextro

44. 5 – butyloctane is an incorrect IUPAC name, give the correct IUPAC name for the Intended compound (a) 3- butyloctane (b) 4- ethyl decane (c) 5- propyl nonane (d) 6-ethyl decane

45. The dehydrogenation of an alkene compound will lead to the formation of an organic compound _____ (a) Alcohol (b) Alkane (c) Alkyne (d) Alkene

46. The process of breaking long chain alkane into smaller molecules is called (a) Breakages (b) Cracking (c) Separation (d) Distillation

47. How many P – orbitals is unaffected during sp hybridization (a) 0 (b) 4 (c) 6 (d) 2

48. The carbon atoms in $CH_3CH=CH_2$ are arranged at (a) 60o (b)120o (c) 124.5o (d) 180o

49. _____ and _____ are amino acids that are predominantly found in the liver and urea cycle (a) Aspartic acid and glycine (b) Citrulline and Ornithrine (c) Serine and praline (d) Arginane and tyrosine

50. Keratin is a _____ protein type (a) Primary (b) Fibrous (c) Conjugated (d) Globular

31. The following are components of natural gas except (a) Methane (b) Butane (c) Propyene

(d) Propane

52. _____ can undergo further reaction with a second equivalent alcohol to yield an acetal (a) Aldehydes (b) Ketone (c) Hemiacetals (d) Alkane

53. _____ are electron rich species that can form a covalent bond by donating two electrons to electron –deficient site. (a) Electrophiles (b) Nucleophiles (c) Aromatics (d) Heterophiles

54. The IUPAC name for the compound below is _____



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(a) Methyl propanoate (b) Ethyl ethanoate (c) carboxylic acid (d) Ethyl propanote

55. Arrange the following functional groups in order of decreasing priority. (a) Carboxylic acid – Ketone – Aldehyde (b) Aldehyde – Ketone – Carboxylic acid (c) Carboxylic acid – Aldehyde – Ketone (d) Ketone – Aldehyde – Carboxylic acid

56. _____ is used for the preservation of biological specimen (a) Tollen (b) Formaline (c) hydrazone (d) Arginine

57. _____ are found in milk, butter & cheese (a) Propanoic acid (b) Ethanoic acid (c) Methanoic acid (d) Benzoic acid

58. Vinyl chloride is use in the manufacturing of the following except (a) Pulp (b) Bottle (c) Rain coat (d) Water pipes

59. Starch can be separated into two principal polysaccharide, which are (a) Amylase & amylopectin ((b) Glycogen & cellulose (c) Amylase & cellulose (d) Glycogen & amylase

60. Benzene undergoes substitution reaction because of (a) Single bonds (b) Delocalized electron

(c) The resonance hybrid (d) The double bonds

61. _____ is among the non – polar side chains (a) Glycine (b) Theonine (c) Serine (d) Proline

62. One of the major property of amino acids is the basicity of _____ group of histidine (a) Inositol (b) Pyruvate (c) Imidazone (d) Acetylco A

63. Amino sugars are denoted by the presence of which group? (a) Ammonium group (b) Ketone group (c) Amine group (d) Aldehyde Group

64. Secondary alcohol is oxidized to _____ (a) Alkanal (b) Alkanone (c) Alkene (d) Alkanoic

Which of the followings are disaccharides? (a) Sucrose, glucose & lactose (b) Maltose, dextrose & sucrose (c) Lactose, sucrose & maltose (d) Dextrose, glucose & lactose

_____ are hydrocarbon that contains aliphatic & aromatic compound (a) Alkenes (b) Arenes (c) Alkane (d) Alcohol

_____ is an example of trihydric alcohol (a) Glycol (b) Alcohol (c) Methanol (d) Glycerol

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_____ is not among the non-polar side chain (a) Alanine (b) Leucine (c) Glutamic acid (d) Phenylalanine

In an Sp^3 hybridization, geometry & angle of inclination are _____ & _____ respectively

(a) Planar / 120° (b) Tetrahedral / 90° (c) Planar / 109° (d) Linear / 180°

The name given to a dissachsride having 2 units of D-glucopyranose joined by a glycosidic bond is _____ (a)Sucrose (b) Maltose (c) Lactose (d) Dextrose